Assignment:3

An insurance company offers two types of insurance products. One protects against medical expenses in case of hospitalisation and the other provides insurance against motor accidents for owners of personal motor cars.

The attached data set provides a sample of 5000 records pertaining to policyholders with the claim amount in case they have filed a claim and 0 in case they have had no claims in the past year.

1. Summarise the data and share key features of both the portfolios
2. Fit appropriate distributions for frequency and severity. Consider splitting the sample into parts to improve the distribution fit if necessary. Comment on your findings
3. Create a sample of 25,000 simulated claims for both the portfolios based on your model from 2. and use this for the subsequent questions
4. From your study of the data and distribution fit can you suggest possible product variants? (example deductible, sub-limits on sum insured etc. with justification)
5. Calculate VaR, TVaR and estimate capital requirements for both the products. Estimate these features for combined distribution and body and tail separately for both the products. Comment on your results
6. Estimate a pure premium for both the products individually.
7. Estimate the premium for the health product with a deductible of 50,000 Rs and motor product with a deductible of Rs. 5000. Comment on your results
8. The insurer wants to reduce his capital requirement and is considering reinsurance. Can you suggest a potential reinsurance transaction which would optimise his capital requirement
9. HARD question. Estimate the diversification effect of offering both the products vs. offering each product individually. (consider modelling a Copula with rho=0.25)
10. Answer question 8. Using the output from your model in 9.